



HEATER FOUNDATIONS

PURPOSE

This document establishes guidelines and recommended procedures for the design of foundations for cylindrical and rectangular heaters.

SCOPE

This document includes the following major sections:

- CYLINDRICAL HEATERS
- SUPPLIER INFORMATION
- DISTRIBUTION OF LOADS
- ESTIMATING PIER AND FOOTING DIMENSIONS
- MINIMUM FOOTING THICKNESS
- FOUNDATION DESIGN
- RECTANGULAR HEATER GUIDELINES
- REFERENCES
- ATTACHMENTS

APPLICATION

This document is to be used for the design of heater foundations and as a guideline for the design of similar combined mats.

CYLINDRICAL HEATERS

The analysis and design of a cylindrical heater foundation with a solid octagonal base on soil or on piles presents a complex problem. Some simplifying assumptions must be made in order to produce an adequate design within a reasonable length of time. For unusually large heaters, another type of base such as annular may be considered more economical than a solid base. The following is the recommended approach to the foundation design.

SUPPLIER INFORMATION

The following information must be provided by the heater supplier. When reviewing supplier drawings for approval, ensure that all of this information has been provided:

- Total empty and operating weight.
- Load taken by center column, if center column provided.



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- Total overturning moment and lateral load from wind and seismic at bottom of heater or bottom of baseplate.
- Size or length of perimeter columns.
- Baseplate, anchor bolt sizes, and diameter of bolt holes.

It is good engineering practice to make a rough check of the overturning moment and lateral load from wind and seismic as provided by the supplier.

It is important to determine if piping flange forces (vertical, horizontal, or both) have been included in the information listed above. Discuss with the Mechanical Engineer and Piping Stress Engineer.

DISTRIBUTION OF LOADS

Vertical Loads

Vertical loads from dead weight are assumed equally distributed to each outer column except where interior columns support a portion of the heater floor.

Vertical loads due to overturning moments are assumed to be distributed to each column in proportion to its distance from the neutral axis of the column group.

Horizontal Loads

Horizontal loads will be distributed to the outer heater columns in the direction of the horizontal force and in proportion to their relative flexural rigidities.

Heater columns are assumed fixed at their base and pinned at the bottom of the floor beams when the columns stop at the floor beams. Columns are assumed pinned at their base and fixed at the bottom of the floor beams when the columns continue up the outside of the heater shell.

ESTIMATING PIER AND FOOTING DIMENSIONS

It is necessary to assume a trial design for the purpose of analysis. Pier sizes can be determined from column baseplate dimensions and minimum edge distances for anchor bolts. Footing size can be accurately estimated by use of the Octagon Bases Sizing Chart for footings on soil. Refer to Structural Engineering Guideline 000.215.1220: Vertical Vessel Foundations. Footings on piles are sized according to heater and pile configuration and spacing limitations. The minimum octagon size equals pier circle plus pier width plus edge distance.



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MINIMUM FOOTING THICKNESS

Footings On Soil

The footing thickness should be adequate to ensure sufficient rigidity for distributing soil bearing pressures in a uniform manner. To ensure sufficient rigidity for the footing, the footing thickness may be determined from the following criteria, using the largest of the conditions:

$$t = 1'-0''$$

$$t = \frac{D'^{*}}{12}$$

$$t = \frac{x-b}{3}$$

* Based on information from K. Terzaghi, Evaluation of Coefficients of Subgrade, Reaction, Geotechnique (London), V.5, December, 1955.

where

t	=	Footing thickness in feet
D'	=	Diameter of pier circle in feet
x	=	Maximum projected distance between adjacent pier centerlines in feet
b	=	Width of concrete pier parallel with 'x' in feet

Footings On Piles

The minimum pile cap thickness above bottom reinforcement will be 1'- 0", according to ACI (American Concrete Institute) 318-05, Section 15.7, with incremental increases of 2 inches.

FOUNDATION DESIGN

Pier Design

Piers will be designed as columns according to ACI-318. Minimize the column width perpendicular to the pier diameter in order to maximize access to the burner floor for piping and personnel. For piers of height less than 3 times the least dimension, the minimum vertical reinforcement can be ½ percent of the gross area of the pier.



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Anchor Bolts

Although the anchor bolts are usually specified by the heater manufacturer, their size and capacity must be checked against the structural specification for the project. The supplier may specify a material other than A307.

The design of anchor bolt embedment, edge distance, and special reinforcement shall be in accordance with Structural Engineering Guideline 000.215.STE05121: Anchor Bolt Design Guide.

Footing Size

For service loads, other than seismic, the minimum stability ratio against overturning and sliding shall be 1.5. If the dead load factor is 0.6 then for above service loads the minimum stability ratio of 1.0 against overturning and sliding shall be permitted. For seismic condition, minimum stability ratio against overturning and sliding shall be 1.0. Load combinations for wind and seismic shall be as specified in 000.215.STC01015 and as amended for specific project. This shall be checked in the direction of the flat side of the octagon. For footings on piles, pile uplift capacities will be checked.

Soil bearing pressure shall be checked against maximum allowable soil bearing on the diagonal of the octagon (at the post of the octagon, rather than the flat). It is acceptable for concrete design of the footing to use soil bearing pressure computed on the flat of the octagon.

When total footing area is not in compression, the soil bearing pressure will be computed using the chart for Octagonal Bases Foundation Pressures, which takes into account the increased soil pressure due to uplift. Refer to vertical vessel foundation guideline 000.215.STE03350.

Foundations on piles should be analyzed according to Structural Engineering Guideline 000.215.1232: Driven Pile Foundations.

Footing Shear And Moment Operating Loads Only

The following formulas are to be used only for octagon foundations on soil. Three formulas are identical to those used for a circular plate, uniformly loaded, with or without a cantilever. Refer to ACI Journal Proceedings, "Strength and Deflection of Circular Uniformly Loaded Slab." Poissons' ratio μ is taken as 0.20 for concrete.

For maximum moment at the piers, M_{Up} , causing bottom tension, use:



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$$M_{Up} = \left[qD^2 \left[-0.075 \log\left(\frac{D}{D'}\right) + 0.025 + \frac{0.025}{\left(\frac{D}{D'}\right)^2} \right] \right] 1.4$$

For maximum moment at the center of the footing, M_{UC} , causing top tension, use:

$$M_{UC} = \left[qD^2 \left[-0.075 \log\left(\frac{D}{D'}\right) + 0.025 - \frac{0.025}{\left(\frac{D}{D'}\right)^2} \right] \right] 1.4$$

This formula for M_{UC} is applicable for the range:

$$1.0 < \frac{D}{D'} < 1.6$$

The moment reverses, producing tension at the bottom when:

$$\frac{D}{D'} > 1.6$$

For values of positive and negative shearing forces, use:

$$V_{U1} = \left[-\frac{qD'}{4} \right] 1.4$$

$$V_{U2} = \left[\frac{-qD'}{4} \left(\frac{D'}{D} - \frac{D}{D'} \right) \right] 1.4$$

where

D	=	Diameter of octagon (inscribed circle) in feet
D'	=	Diameter of pier (inscribed circle) in feet
q	=	Net soil bearing, kip/ft ²
V_{U1}	=	Shearing force per unit length, inside the pier circle
V_{U2}	=	Shearing force per unit length, outside the pier circle

Note!!! The above formulae are derived for the case where there is no center pier. However, they



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may be used for the case when a center pier is required because the values of shearing force and moment are on the conservative side.

Empty And Operating Loads In Combination With Lateral Loads

For foundations supported on soil or piles, the procedure described here is used to compute the shearing force and moment values.

The footing analysis for empty or operating conditions in combination with lateral forces is performed by transferring all pier loads and moments to the center 1'- 0" wide strip. When making this approximation, the resulting free-body will not be exactly in balance because of the octagonal shape of the footing. This condition is corrected by the adjustment of vertical unit loads.

- Find a fictitious soil bearing or the fictitious pile reactions for the entire footing using factored loads.
- Divide each load and moment by the width of the footing at that point to get unit loads and moments.
- Using these unit loads and the total soil bearing value or pile reactions, make a free body of the 1 foot wide center strip of the footing.
- Balance this free body by modifying the vertical unit loads thus:
- Vertical Unit Load Modification = $\frac{P}{N} \pm \frac{M_c}{I}$

where

P = Unbalanced vertical load, kips

N = Number of vertical load lines on free body

M = Unbalanced moment about centerline of free body, foot kips

I = Moment inertia of load lines about centerline = $\sum c^2$

c = Distance from centerline to vertical unit loads

Beam Shear and peripheral (punching) shear are checked to determine whether the footing thickness is adequate. Maximum value of the shearing force from all the loading cases is used to check the beam shear. The shear stresses on the critical periphery (punching shear) surrounding the pier must be investigated as stated in ACI 318-05, Sections 11.12.2 and 11.12.6.

Footing Reinforcing

- The maximum amount of reinforcement per foot determined for top steel and bottom steel, respectively, will be placed uniformly over the entire footing area in two



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- orthogonal directions.
- Care must be taken to ensure that the bottom reinforcement can be developed between the face of the pier and the edge of the footing.

RECTANGULAR HEATER GUIDELINES

Distribution Of Vertical And Horizontal Loads

The distribution of vertical and horizontal loads must be provided by the heater supplier. This load distribution is dependent on the internal structure of the heater, which dictates how vertical and horizontal loads pass through the structure to the concrete piers. Vertical and horizontal loads and moments should be provided (usually in tabular form) at each pier, for each load case. The engineer should verify the reasonableness of net overturning and base shear as well as distribution of loads from the supplier.

Mat Foundation Vs Individual Spread Footings

Mat foundations are commonly used to support the many columns extending below a rectangular heater. The following 2 issues should be considered when comparing a mat foundation to individual spread footings for each pier:

- Differential settlement**
If soil conditions are such that differential settlement may occur among the various piers, and if this settlement would damage the heater, choose a mat foundation.
- Solidity ratio**
This is the product of the area of the mat divided by the total area of all individual spread footings. The installed unit cost (dollars per cubic yard) of concrete for individual spread footings is about 1.5 to 2 times that of a mat foundation. Assuming equal footing, mat thickness, and excavation costs, if the solidity ratio is greater than 2, individual spread footings would appear to be less expensive. This is not, however, an absolute rule but rather an aid in the decision process.

Location of CG (Center Of Gravity) of Mat Foundation

Generally, it is not necessary to center the mat under the CG of the heater operating loads. Operating load soil pressures under a large mat are usually very low and, therefore, differential settlement is negligible.

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REFERENCES

ACI (American Concrete Institute) 318-05

ACI Journal Proceedings. Strength and Deflection of Circular Uniformly Loaded Slab, Supported Between Center and Periphery. Title Number 60-18. February, 1963.

Terzaghi, K. Evaluation of Coefficients of Subgrade Reaction. Geotechnique (London), V.5. December, 1955.

Structural Engineering
Guideline 000.215.STC01015: Structural Design Criteria

Structural Engineering
Guideline 000.215.STE05121: Anchor Bolt Design Guide

Structural Engineering
Guideline 000.215.1215: Wind Load Calculations

Structural Engineering
Guideline 000.215.1216: Earthquake Engineering

Structural Engineering
Guideline 000.215.STE03350: Vertical Vessel Foundations

Structural Engineering
Guideline 000.215.1232: Driven Pile Foundations

ATTACHMENTS

Attachment 01:

Sample Design 1: Cylindrical Heater Foundation

Note!!! Calculations for this attachment have not been updated to meet the requirements of the latest references and design codes.

Attachment 02:

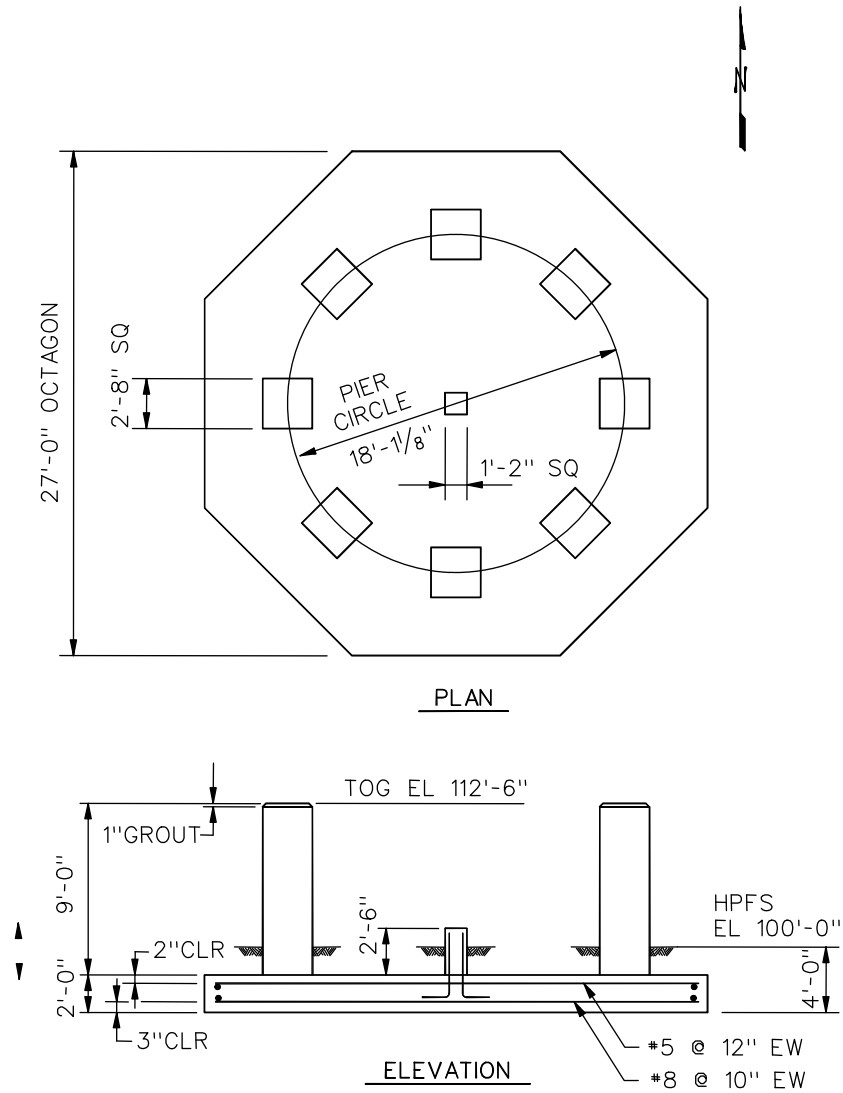
Sample Design 2: Rectangular Heater Foundation

Note!!! Calculations for this attachment have not been updated to meet the requirements of the latest references and design codes.



HEATER CALCULATIONS

Sample Design 1: Cylindrical Heater Foundation



Design Conditions

Soil bearing: 5000 PSF @ 4'-0" below grade

Stability Ratio: $\geq 1.5, 1/3$ Allowable increase for wind & seismic



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Sample Design 1: Cylindrical Heater Foundation

$\gamma_{soil} = 100 \text{ PCF}$ $\gamma_{conc} = 150 \text{ PCF}$
 $f_c = 4000 \text{ PSI}$ $f_y = 60000 \text{ PSI}$
 Anchor Bolt Material: A-307
 Wind Pressure on Concrete Columns: 34 PSF
 Concrete Code: ACI 318-05

Heater Data (From Vendor Drawings)

Columns Exterior: (8) - W10x49 on $18'-1\frac{1}{8}"$ circle

Exterior: Distance from heater bottom to bottom of base plate = 2 ft. (Assume pinned at top and fixed at bottom) Base Plates and Anchor Bolts

Exterior Columns: $1'-10" \times 1'-10" \times 1\frac{1}{2}"$ w/ (4) - $1\frac{3}{4}" \phi$ A.B.'s @ 16" c.c

Interior Columns: $8\frac{1}{2}" \times 8\frac{1}{2}" \times 1\frac{1}{2}"$ w/ 2 - $\frac{3}{4}" \phi$ A.B.'s @ 6" c.c

Vertical Loads (Working)

Empty	Exterior Columns	62.3 K each
	Interior Columns	9.2 K
	Total Weight	507.6 K
Operating	Exterior Columns	66.3 K
	Interior Columns	9.5 K
	Total Weight	539.9 K

Horizontal Loads (@ Bottom of Floor Beams)

Wind Shear	$V_w =$	53.5 K
Wind Moment	$M_w =$	3956.2 K-FT



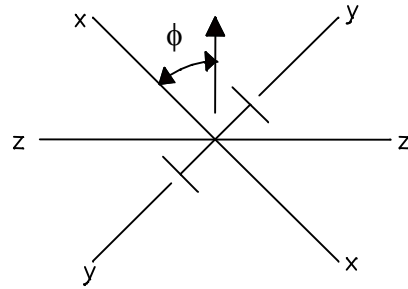
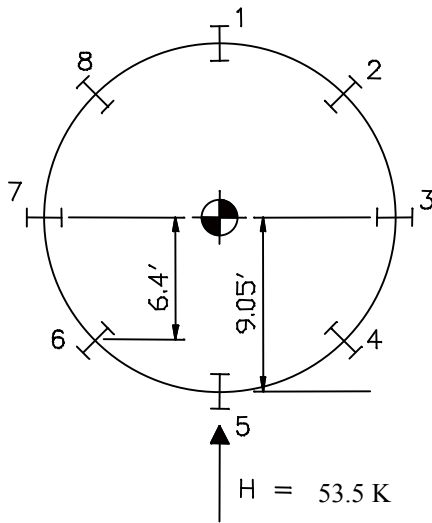
HEATER CALCULATIONS

Sample Design 1: Cylindrical Heater Foundation

SOLUTION:

Horizontal Load Distribution to Exterior Columns

W10x49 $I_x = 272 \text{ in}^4$ $I_y = 93.4 \text{ in}^4$



$$I_z = I_x \sin^2 \phi + I_y \cos^2 \phi \quad (\text{Reference AISC 13th ED. p 17-42})$$

ϕ = Angle between horizontal force and x-x axis of column

Moments of Inertia

$$\phi_{1,5} = 90 \quad I = I_x = 2 @ 272 \text{ in}^4 = 544.0 \text{ in}^4$$

$$\phi_{2,4,6,8} = 45 \quad I = (272) \sin^2 (45) + (93.4) \cos^2 (45) = 136.0 + 46.7 = 182.7 \text{ in}^4$$

$$I = 182.7 \text{ in}^4 \times 4 = 730.8 \text{ in}^4$$

$$\phi_{3,7} = 0 \quad I = I_y = 2 @ 93.4 \text{ in}^4 = 186.8 \text{ in}^4$$

$$\Sigma I = 1461.6 \text{ in}^4$$

Column Shear Distribution

$$V_{\text{col}} = H \left(\frac{I}{\Sigma I} \right)$$



HEATER CALCULATIONS

Sample Design 1: Cylindrical Heater Foundation

Columns 1 & 5

$$V_{1,5} = 53.5 \text{ K} (272 / 1461.6) = 9.95 \text{ K/col}$$

Columns 2, 4, 6 & 8

$$V_{2,4,6 \& 8} = 53.5 \text{ K} (182.7 / 1461.6) = 6.72 \text{ K/col}$$

Columns 3 & 7

$$V_{3,7} = 53.5 \text{ K} (93.4 / 1461.6) = 3.4 \text{ K/col}$$

Vertical Load Distribution $P = (W / n) + (Mc / I)$ $I = nd^2 / 8 = 8 (18.1)^2 / 8 = 327.6 \text{ FT}^2$

Empty

$$P_{1,5} = 62.3 \text{ K} \pm 3956.2 \text{ K-FT} (9.05') / 327.6 \text{ FT}^2 = 62.3 \pm 109.3 = 171.5 \text{ K (Comp) @ } P_1$$

$$47.0 \text{ K (Ten) @ } P_5$$

$$P_{2,4,6,8} = 62.3 \text{ K} \pm 3956.2 \text{ K-FT} (6.40') / 327.6 \text{ FT}^2 = 62.3 \text{ K} \pm 77.26 \text{ K} = 139.6 \text{ K (Comp) @ } P_{2,8}$$

$$14.97 \text{ K (Ten) @ } P_{4,6}$$

$$P_{3,7} = 63.3 \text{ K} \pm 0.00 = 62.3 \text{ K}$$

Operating

$$P_{1,5} = 66.3 \text{ K} \pm 109.3 = 175.6 \text{ K (Comp) @ } P_1$$

$$43.0 \text{ K (Ten) @ } P_5$$

$$P_{2,4,6,8} = 66.3 \text{ K} \pm 77.3 \text{ K} = 143.6 \text{ K (Comp) @ } P_{2,8}$$

$$10.96 \text{ K (Ten) @ } P_{4,6}$$

$$P_{3,7} = 66.3 \text{ K} \pm 0.00 = 66.3 \text{ K}$$

Tensile and Shear Capacity in Anchor Bolt Material (Check per Structural Engineering Guideline 000.215.STE05121 and notations per ACI 318-05)

Tension (with factored loads)

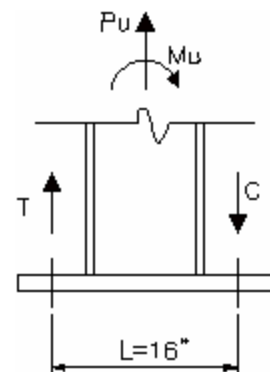
$$P_u (\text{empty}) = (0.9D + 1.6W) = 0.9 \times 62.3 + 1.6 \times 109.3 = 118.8 \text{ K}$$

$$M_u = 2' (9.95 \text{ K}) \times 1.6 = 31.8 \text{ K-FT}$$

$$N_{ua} = \frac{P_u}{4} + \frac{M_u}{2L} = 149.7 \text{ K} / 4 + 31.8 \text{ KFT} (12" / \text{FT}) / 2(16") \text{ per bolt}$$

$$N_{ua} = 37.4 + 11.93 = 49.33 \text{ K}$$

Ref. ACI 318 Appendix D:





HEATER CALCULATIONS

Sample Design 1: Cylindrical Heater Foundation

$$\phi N_n = \phi A_{se} f_{uta} \dots\dots\dots \text{ACI Eq. D-3}$$

$$\phi = 0.75$$

A_{se} = tensile stress area per AISC Table 7-18 = 1.90 in²

F_{uta} = specified tensile strength of anchor steel = 60 ksi for A307

$$\phi N_n = 0.75 \times 1.90 \times 60 = 85.5K$$

Shear

$$V_{ua} = 9.95K / 4 \times 1.6 = 3.98K \quad \text{per bolt}$$

$$\phi V_n = \phi \times 0.6 A_{se} f_{uta}$$

$$\phi V_n = 0.6 \times 0.6 \times 1.9 \times 60 = 41.0K$$

Since A.B. are installed with grout between the base plate and concrete pier ϕV_n is modified by a factor of 0.8 (ACI D6.1.3)

$$\text{Therefore, } \phi V_n = 0.8 \times 41.0K = 32.8K$$

Note: When tension and shear are due to seismic in regions of moderate to high risk, ACI D3.3.3 requires ϕN_n and ϕV_n to be multiplied by 0.75 factor

$$\frac{V_{ua}}{\phi V_n} = \frac{3.98}{32.8} = 0.12 < 0.2$$

Per ACI D7.1, $\phi N_n \geq N_{ua}$ is permitted and no interaction of tensile and shear forces is required.

$$\text{Hence } \phi N_n = 85.5K \geq N_{ua} = 49.33K \quad \text{OK}$$

USE: 4 - 1 3/4 ϕ A307 Anchor Bolts

Note!!! For Brevity, anchor bolt design with respect to concrete anchorage not shown here. See Structural Engineering Guideline 000.215.STE05121, "Anchor Bolt Design Guide".

Minimum Pier Size (Size requirement per Structural Engineering Guideline 000.215.STE05121)

Exterior: Minimum size = 16" c.c. + 2 (4 x 1 3/4 ϕ Bolts) = 32"

TRY 32" x 32" exterior pier, 9'-0" high (assuming 2'-0" footing thickness)

$$Wt = 2.67^2 \times 9 \times 0.15 = 9.60 K$$



HEATER CALCULATIONS

Sample Design 1: Cylindrical Heater Foundation

Interior: Minimum size = 6" c.c. + 2 (4" min edge) = 14"
TRY 14" x 14" interior pier, 2'-6" high

Ultimate Load & Moment with Eccentricity

1. Column #1 (Compression)

Load Condition: Operating + Wind

$$\begin{aligned} P \text{ (Operating + Wind)} &= (66.3 \text{ K} + 9.6 \text{ K}) \text{ D.L.} + (109.3) \text{ K W.L.} \\ M @ \text{ Column Base} &= 11' (9.95 \text{ K}) + 2.67' (7') (0.04 \text{ KSF}) (7' / 2 + 2') \\ &= 109.3 \text{ K-FT} + 4.1 \text{ K-FT} = 113.4 \text{ K-FT} \end{aligned}$$

w/ load factors

$$\begin{aligned} P_u &= 1.2 (75.9 \text{ K}) + 1.6 (109.3 \text{ K}) = 266 \text{ K} \\ M_u &= 1.6 (113.6 \text{ K-FT}) = 181.8 \text{ K-FT} & e = 0.68' \text{ (per ACI Equation 9-4)} \\ P_u &= 0.90 (75.9 \text{ K}) + 1.6 (109.3 \text{ K}) = 243.2 \text{ K} \\ M_u &= 1.6 (113.6 \text{ K-FT}) = 181.8 \text{ K-FT} & e = 0.75' \text{ (per ACI Equation 9-6)} \end{aligned}$$

Load Condition: Empty + Wind

$$\begin{aligned} P \text{ (Empty + Wind)} &= (62.3 \text{ K} + 9.6 \text{ K}) \text{ D.L.} + (128.6 \text{ K}) \text{ W.L.} \\ \text{w/ load factors} \\ P_u &= 1.2 (71.9 \text{ K}) + 1.6 (109.3 \text{ K}) = 261.2 \text{ K} \\ M_u &= 181.8 \text{ K-FT} & e = 0.696' \\ P_u &= 0.90 (71.9 \text{ K}) + 1.6 (109.3 \text{ K}) = 239.6 \text{ K} \\ M_u &= 181.8 \text{ K-FT} & e = 0.758' \end{aligned}$$

2. Column #5 (Uplift Side)

Load Condition: Operating + Wind

$$\begin{aligned} P_u &= 1.2 (75.9 \text{ K}) - 1.6 (109.3 \text{ K}) = -83.8 \text{ K} \\ M_u &= 181.8 \text{ K-FT} & e = 2.17' \\ P_u &= 0.90 (75.9 \text{ K}) - 1.6 (109.3 \text{ K}) = -106.6 \text{ K} \\ M_u &= 181.8 \text{ K-FT} & e = 1.71' \end{aligned}$$



HEATER CALCULATIONS

Sample Design 1: Cylindrical Heater Foundation

Load Condition: Empty + Wind

$$P_u = 1.2 (71.9 \text{ K}) - 1.6(109.3 \text{ K}) = -88.6 \text{ K}$$

$$M_u = 181.8 \quad e = 2.05'$$

$$P_u = 0.90 (71.9 \text{ K}) - 1.6(109.3 \text{ K}) = -110.2 \text{ K}$$

$$M_u = 181.8 \text{ K-FT} \quad e = 1.65'$$

Design Columns Subject to Biaxial Bending (Cols 2, 4, 6, 8)

$$P_{\text{empty}} = 71.9 \text{ K} \pm 77.3 \text{ K} \quad P_{\text{operating}} = 75.9 \text{ K} \pm 77.3 \text{ K}$$

$$H = 6.72 \text{ K/col} @ 45^\circ \text{ to the x-x axis of the column}$$

$$H_x = 0.707 (6.72 \text{ K/col}) = 4.75 \text{ K} \quad H_y = 4.75 \text{ K}$$

$$M_x = M_y = 11' (4.75 \text{ K}) = 52.3 \text{ K-FT} + \text{Wind Moment}$$

$$= 52.3 \text{ K-FT} + 0.707 (2.67') (7') (0.034 \text{ KSF}) (7' / 2 + 2') (\sqrt{2}) = 55.8 \text{ K-FT}$$

3. Columns #2, #8 (Compression Side)

Load Condition: Operating + Wind

$$P_u = 1.2 (75.9 \text{ K}) + 1.6 (77.3 \text{ K}) = 214.8 \text{ K}$$

$$M_{ux} = M_{uy} = 55.8 \text{ K-FT} (1.6) = 89.3 \text{ K-FT} \quad e = 0.42'$$

$$P_u = 0.90 (75.9 \text{ K}) + 1.6 (77.3 \text{ K}) = 192.0 \text{ K}$$

$$M_{ux} = M_{uy} = 55.8 \text{ K-FT} (1.6) = 89.3 \text{ K-FT} \quad e = 0.465'$$

Load Condition: Empty + Wind

$$P_u = 1.2 (71.9 \text{ K}) + 1.6 (77.3 \text{ K}) = 209.96 \text{ K}$$

$$M_{ux} = M_{uy} = 55.8 \text{ K-FT} (1.6) = 89.3 \text{ K-FT} \quad e = 0.425'$$

$$P_u = 0.90 (71.9 \text{ K}) + 1.6 (77.3 \text{ K}) = 188.4 \text{ K}$$

$$M_{ux} = M_{uy} = 55.8 \text{ K-FT} (1.6) = 89.3 \text{ K-FT} \quad e = 0.474'$$

4. Columns #4, #6 (Uplift Side)

Load Condition: Operating + Wind

$$P_u = 1.2 (75.9 \text{ K}) - 1.6 (77.3 \text{ K}) = -32.6 \text{ K}$$

$$M_{ux} = M_{uy} = 89.3 \text{ K-FT} \quad e = 2.74'$$



HEATER CALCULATIONS

Sample Design 1: Cylindrical Heater Foundation

$$P_u = 0.90 (75.9 \text{ K}) - 1.6 (77.3 \text{ K}) = -55.4 \text{ K}$$

$$M_{ux} = M_{uy} = 89.3 \text{ K-FT} \quad e = 1.61'$$

Load Condition: Empty + Wind

$$P_u = 1.2 (71.9 \text{ K}) - 1.6 (77.3 \text{ K}) = -37.4 \text{ K}$$

$$M_{ux} = M_{uy} = 89.3 \text{ K-FT} \quad e = 2.38'$$

$$P_u = 0.90 (71.9 \text{ K}) - 1.6 (77.3 \text{ K}) = -58.97 \text{ K}$$

$$M_{ux} = M_{uy} = 89.3 \text{ K-FT} \quad e = 1.51'$$

Column Design

Note!!! For brevity, concrete column design not shown here.

Footing Analysis and Design

Minimum octagon size is pier circle + pier width + edge distance = $(18' - 1\frac{1}{8}" + 32" + 2(6")) = 21' - 9\frac{1}{8}"$

TRY D = 22'-0"

Minimum Thickness Criteria

1. $t \geq 12"$
2. $t = P.C. / 12 = 18.09' / 12 = 1.51'$
3. $t = (x^* - b) / 3 = (6.4' - 2.69') / 3 = 1.24' \quad x^* = 18.09 / 2 \sin(360 / 8) = 6.4'$
4. t must be sufficient to develop column steel

For #9 bars:

Development of standard hooks in tension (ACI 318-05, Section 12.5)

$$l_{dh} = \frac{1200d_b}{\sqrt{f'_c}} = \frac{1200 \times (1.128')}{\sqrt{4000}} = 21.4" \quad (0.02\psi_e\lambda f_y / f'_c) d_b \text{ with } \psi_e \text{ and } \lambda = 1.00$$

(ACI 12.5.2)

$$l_{dh} = 0.7 \times 21.4" = 15.0" \quad (\text{ACI 12.5.3(a)})$$

$$t = 15.0" + 2(1") + 3" = 20.0"$$

5. t must be great enough to resist shear & torsion in footing



HEATER CALCULATIONS

Sample Design 1: Cylindrical Heater Foundation

Therefore, TRY 22'-0" octagon 2'-0" thick ($A = 401.0 \text{ FT}^2$)

Check Soil Bearing (Allowable Stress)

Octagon Size: 22'-0" (2'-0" Thick)

Wt. of exterior cols = $8 \times (2.67')^2 \times 9.0' \times 0.15 \text{ KCF} = 77.0 \text{ K}$

Wt. of interior cols = $1 \times (1.17')^2 \times 2.5' \times 0.15 \text{ KCF} = 0.51 \text{ K}$

Wt. of footing = $401.0 \text{ FT}^2 \times 2' \times 0.15 \text{ KCF} = 120.3 \text{ K}$

Wt. of soil above footing = $2.0 (401.0 - 8 (2.67')^2 - 1 (1.17')^2) \times 0.10 \text{ KSF}$
 $= 2.0 (401.0 - 57.0 - 1.4) \times 0.10 \text{ KCF} = 68.5 \text{ K}$

Total Weight = 266.3 K

Heater Weights Empty = 507.6 K Operating = 539.9 K

$P_{\text{empty}} = 507.6 \text{ K} + 266.3 \text{ K} = 774.0 \text{ K}$

$P_{\text{operating}} = 539.9 \text{ K} + 266.3 \text{ K} = 806.2 \text{ K}$

Moment @ ftg base = $3956.2 \text{ K-FT} + 13.0' (53.5 \text{ K}) = 4651.2 \text{ K-FT}$

Empty + Wind

$e = M / P = 4651.2 \text{ K-FT} / 774.0 \text{ K} = 6.0'$ $e / D = 6.0' / 22.0' = 0.27^*$

* Refer to octagon chart in Fig. B Guideline 000.215.STE 03350

$L = 3.8$ $SB = LP / A = 3.8 (774.0 \text{ K}) / 401.0 \text{ FT}^2 = 7.33 \text{ KSF} > 5.0 \text{ KSF} \times 1.33 = 6.65 \text{ KSF}$

Check Soil Bearing for 27'-0" (2'-0" thick)

Wt. of columns = 77.5 K

Wt. of footing (603.9 FT^2) (2') (0.15 KCF) = 181.2 K

Wt. of soil ($603.9 - 58.4$) (2') (0.10 KCF) = 109.1 K

Weight of foundation = 367.8 K

$P_{\text{empty}} = 507.6 \text{ K} + 367.8 \text{ K} = 875.4 \text{ K}$

$P_{\text{operating}} = 539.9 \text{ K} + 367.8 \text{ K} = 907.7 \text{ K}$

Moment @ ftg base = 4651.2 K-FT

Empty

$e = M / P_{\text{emp}} = 4651.2 \text{ K-FT} / 875.4 \text{ K} = 5.31'$ $e / D = 5.31' / 27.0' = 0.197$



HEATER CALCULATIONS

Sample Design 1: Cylindrical Heater Foundation

$$L_{diag} = 2.65 \quad SB = LP_{emp} / A = 2.65 (875.4 \text{ K}) / 603.9 \text{ FT}^2 = 3.84 \text{ KSF} < 5.0 \text{ KSF} \times 1.33 = 6.65 \text{ KSF}$$

Operating

$$e = M / P_{oper} = 4651.2 \text{ K-FT} / 907.7 \text{ K} = 5.12' \quad e / D = 5.12' / 27.0' = 0.19$$

$$L_{diag} = 2.54 \quad SB = LP_{oper} / A = 2.54 (907.7 \text{ K}) / 603.9 \text{ FT}^2 = 3.82 \text{ KSF} < 5.0 \text{ KSF} \times 1.33 = 6.65 \text{ KSF}$$

Stability Check (Empty + Wind)

$$SR = D / 2 (e) = 27.0' / 2 (6.25') = 2.16 > 1.5 \quad \text{O.K.}$$

USE: 27'-0" Octagon

Check footing thickness for punching shear (Reference ACI 318-05, Sections 11.12.2 & 11.12.6)

From column analysis (operating + wind @ column 1):

$$V_u = P_u = 1.2 (66.3 \text{ K} + 9.6 \text{ K}) + 1.6 (109.3 \text{ K}) = 266.0 \text{ K}$$

$$\begin{aligned} M @ \text{column base} &= 11' (9.945 \text{ K}) + 2.67' (7') (0.034 \text{ KSF}) (7' / 2 + 2') \\ &= 109.3 \text{ K-FT} + 3.5 \text{ K-FT} = 112.8 \text{ K-FT} \end{aligned}$$

$$M_u = 1.6 (112.8 \text{ K-FT}) = 180.5 \text{ K-FT}$$

$$\text{Pier circle, PC} = 18.1' \quad \text{Octagon diameter, D} = 27'-0"$$

l = distance from edge of mat to edge of column

$$l = (27.0' - 18.1') / 2 - (2.67' / 2)$$

$$l = 4.5' - 1.3' = 3.2' = 38.4"$$

$$d = 24" - 3" - 1" = 20"$$

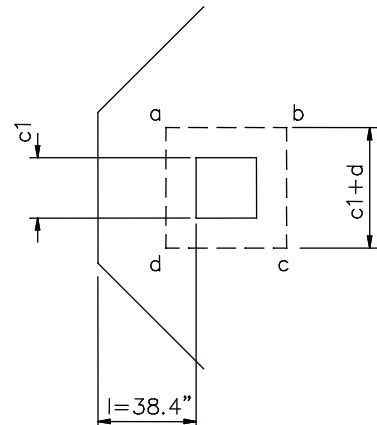
$$\text{Perimeter a-b-c-d} = 4 (c_1 + d) = 4 (32" + 20") = 208.0"$$

Moment transfer in slab-column connections

$$V_u = \frac{V_u}{A_c} \pm \frac{\gamma_v M_u c_{AB}}{J_c} \quad (\text{ACI 318-05, Section R11.12.6.2})$$

$$c_{AB} = (c_1 + d) / 2 = (32" + 20") / 2 = 26"$$

$$A_c = 208.0" (d) = 208.0" (20") = 4160.0 \text{ in}^2$$





HEATER CALCULATIONS

Sample Design 1: Cylindrical Heater Foundation

$$J_c = \frac{d(c_1 + d)^3}{6} + \frac{(c_1 + d)d^3}{6} + \frac{d(c_2 + d)(c_1 + d)^2}{2}$$

$$= \frac{20(52)^3}{6} + \frac{52(20)^3}{6} + \frac{20(52)^3}{2}$$

$$= 468,693.3 + 69,333.3 + 1,406,080.0$$

$$J_c = 1.94 \times 10^6$$

$$\gamma_v = 1 - \gamma_f \quad (\text{ACI 11 - 39}) \quad \text{Where} \quad \gamma_f = \frac{1}{1 + \left(\frac{2}{3}\right)\sqrt{\frac{b_1}{b_2}}} \quad b_1 = b_2, \gamma_v = 0.40$$

$$V_u = \left\{ \frac{297 K}{4160 .0 in^3} \pm \frac{0.40 (212 .48 K - FT)(12'' / FT)(26'')}{1.94 \times 10^6 in^4} \right\} \left(\frac{1}{0.85} \right)$$

$$V_u = (0.0714 \pm 0.0137) \left(\frac{1}{0.85} \right) = 0.100 \leq (0.85)4\sqrt{4000} = 0.215 \quad \text{O.K.}$$

Design of footing reinforcement

Operating + Wind

$$P = 907.7 K \quad M = 4651.2 K\text{-FT}$$

Required Strength, U

$$1. \quad U = 1.2 D + 1.6 W \quad (\text{Reference ACI 318-05, Section 9-2})$$

$$P_u = 1.2 (907.7 K) = 1089.2 K$$

$$M_u = 1.6 (4651.2 K\text{-FT}) = 7442 K\text{-FT}$$

$$e = M_u / P_u = 7442 K\text{-FT} / 1089.2 K = 6.83'$$

$$e / D = 6.83 / 27.0' = 0.25$$

$$L_{flat} = 4.5 \quad k = 0.49$$

$$SB = L P_u / A = (4.5)(1089.2 K) / 603.9 FT^2 = 8.17 KSF$$



HEATER CALCULATIONS

Sample Design 1: Cylindrical Heater Foundation

$$kD = (0.49)(27.0') = 13.23'$$

2. $U = 0.90 D + 1.6 W$

$$P_u = 0.90 (907.7 K) = 816.9 K$$

$$M_u = 1.6 (4651.2 K-FT) = 7442 K-FT$$

$$e = M_u / P_u = 7442 K-FT / 816.9 K = 9.1'$$

$$e / D = 9.1' / 27.0' = 0.34$$

$$L_{flat} = 6.0 \quad k = 0.61$$

$$SB = L P_u / A = (6)(816.9 K) / 603.9 FT^2 = 8.12 KSF$$

$$kD = (0.61)(27.0') = 16.5$$

Empty + Wind

$$P = 875.4 K \quad M = 4651.2 K-FT$$

Required Strength, U

1. $U = 1.2 D + 1.6 W$ (Reference ACI 318-05, Section 9-2)

$$P_u = 1.2 (875.4 K) = 919.2 K$$

$$M_u = 1.6 (4651.2 K-FT) = 7442 K-FT$$

$$e = M_u / P_u = 7442 K-FT / 919.2 K = 8.1'$$

$$e / D = 8.1' / 27.0' = 0.3$$

$$L_{flat} = 4.50 \quad k = 0.4935$$

$$SB = L P_u / A = (4.5)(919.2 K) / 603.9 FT^2 = 6.85 KSF$$

$$kD = (0.4935)(27.0') = 13.32'$$

2. $U = 0.90 D + 1.6 W$

$$P_u = 0.90 (875.4 K) = 787.9 K$$

$$M_u = 1.6 (4651.2 K-FT) = 7442 K-FT$$



HEATER CALCULATIONS

Sample Design 1: Cylindrical Heater Foundation

$$e = M_u / P_u = 7442 \text{ K-FT} / 787.9 \text{ K} = 9.45$$

$$e / D = 9.45 / 27.0' = 0.35$$

$$L_{\text{flat}} = 6.55 \quad k = 0.62$$

$$SB = L P_u / A = (6.55)(787.9 \text{ K}) / 603.9 \text{ FT}^2 = 8.55 \text{ KSF}$$

$$kD = (0.62)(27.0') = 16.74'$$

Consider Empty + Wind

Distributed load due to soil and footing

$$0.9 P / A = 0.9 (181.2 \text{ K} + 109.1 \text{ K}) / 603.9 \text{ FT}^2 = 0.43 \text{ KSF}$$

Evaluation of unit loads

$$U = 0.9 D + 1.6 W$$

$$P_1 = \frac{0.9(62.3 \text{ K} + 9.6 \text{ K}) + 1.6(109.4 \text{ K})}{20.0'} = \frac{64.7 + 175}{20.0'} = 12 \text{ K}$$

$$P_2 = P_8 = \frac{2[0.9(71.9 \text{ K}) + 1.6(77.3 \text{ K})]}{26.0'} = \frac{2(64.71 + 123.6)}{26.0'} = 14.5 \text{ K}$$

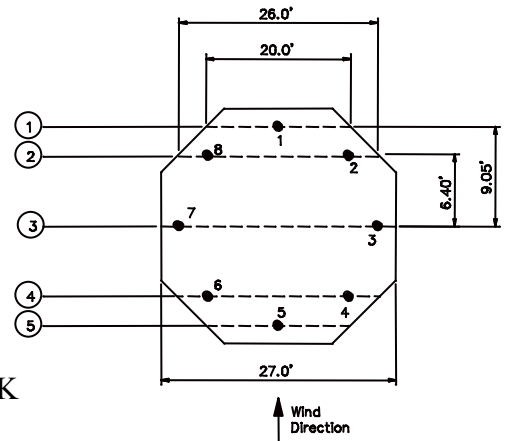
$$P_{3,7} \text{ \& int_Col} = \frac{2(0.9)(71.9 \text{ K}) + 0.9(9.2 \text{ K} + 1.6 \text{ K})}{27.0'}$$

$$= \frac{129.42 + 9.72}{27.0'} = 5.15 \text{ K}$$

$$P_{4,6} = \frac{2[(0.9)(71.9 \text{ K}) - 1.6(77.3 \text{ K})]}{26.0'} = \frac{129.42 - 247.2 \text{ K}}{26.0'} = -4.53 \text{ K}$$

$$P_5 = \frac{0.9(71.9 \text{ K}) - 1.6(109.3 \text{ K})}{20.0'} = \frac{64.71 - 174.9}{20.0'} = -5.51 \text{ K}$$

$$M_1 = \frac{(1.6)(9.945 \text{ K / col})(13.0')}{20.0'} = 10.3 \text{ K' / ft}$$





HEATER CALCULATIONS

Sample Design 1: Cylindrical Heater Foundation

$$M_{2,8} = \frac{(1.6)(2)(6.71K)(13.0')}{26.0'} = 10.74K' / ft$$

$$M_{3,7} = \frac{(1.6)(2)(3.4)(13.0')}{27.0'} = 5.24K' / ft$$

$$M_{4,6} = 10.74K' / ft$$

$$M_5 = 10.3K' / ft$$

Evaluation of unbalanced loads & moments

$$\Sigma P = S.B(\uparrow) - \text{Soil / ftg. load}(\downarrow) - \text{col. loads}(\downarrow) + \text{col. loads}(\uparrow)$$

$$\Sigma P = \frac{1}{2}(8.55)(27.0' - 16.8') - 0.43(27.0') - 12.0K - 14.5K - 5.15K + 4.53K + 5.51K$$

$$\Sigma P = 43.6K - 11.6K - 21.6K$$

$$\Sigma P = 10.4K$$

Take ΣM about centerline (clockwise = positive moment)

$$\Sigma M = 43.6K \left(\frac{27.0}{2} - \frac{10.3}{3} \right) - (12K + 5.51K)(9.05') - (14.5K + 4.53K)(6.40') - 2(10.3K') - 2(10.74K') - 5.2K'$$

$$\Sigma M = 438.9 - 158.5 - 121.8 - 20.6 - 21.5 - 5.2$$

$$\Sigma M = 111.3K'$$

$$\text{Load Modification} \left(\frac{P}{N} \pm \frac{Mc}{I} \right)$$

P = Unbalanced vertical load

N = Number of load lines

M = Unbalanced moment

$$I = \Sigma c^2 \quad c = \text{Distance from N.A. to load line}$$

$$I = \Sigma c^2 = 2(9.05')^2 + 2(6.4')^2 = 245.7$$

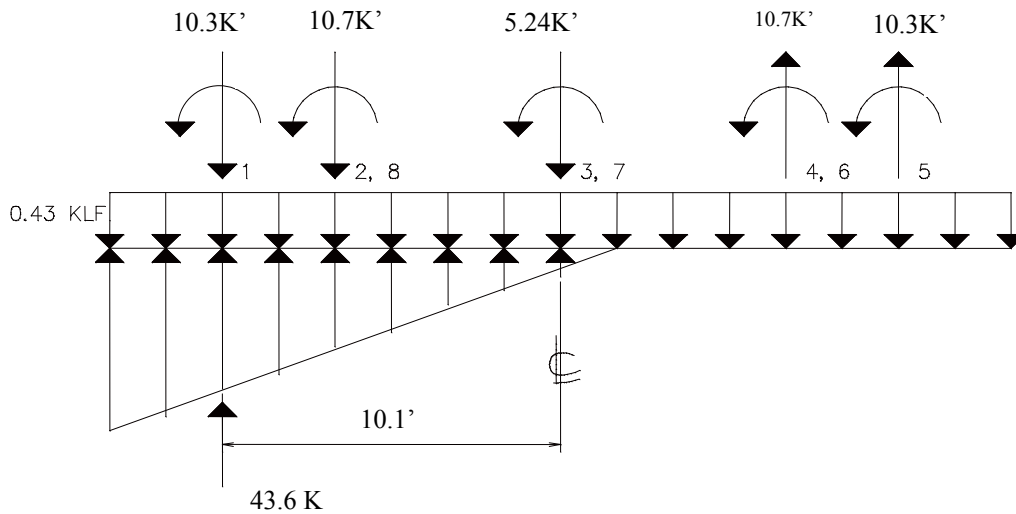
$$\text{@column line 1.} \quad = \frac{10.4}{5} + \frac{(111.3)(9.05)}{245.7} = 2.08 + 4.1 = 6.18K$$



HEATER CALCULATIONS

Sample Design 1: Cylindrical Heater Foundation

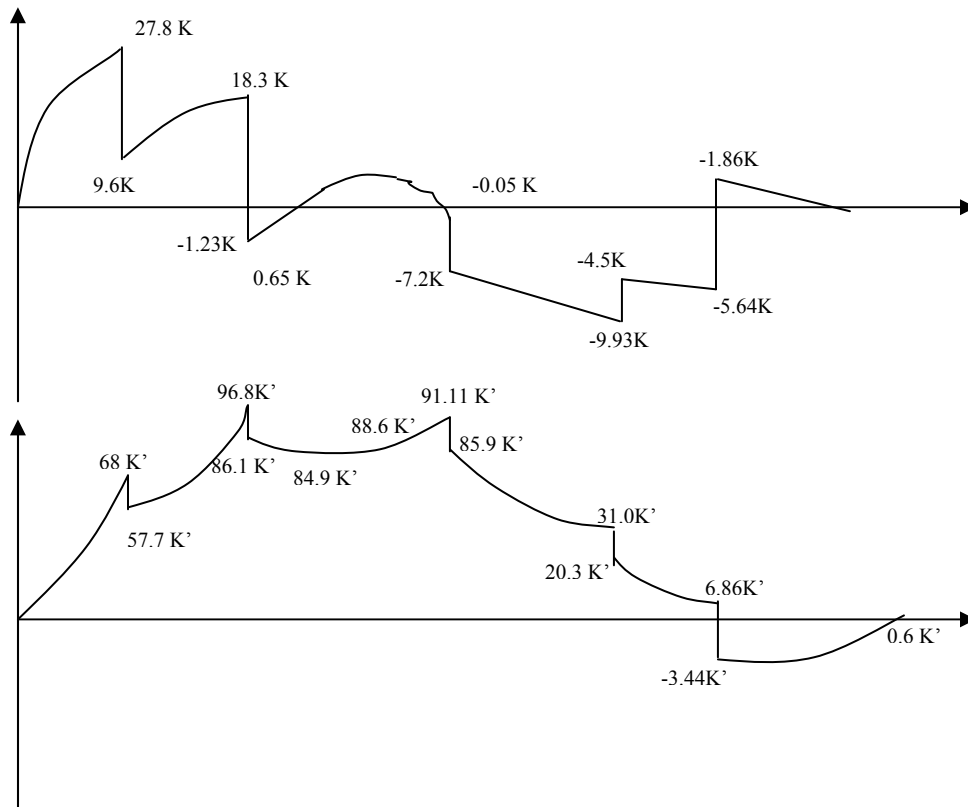
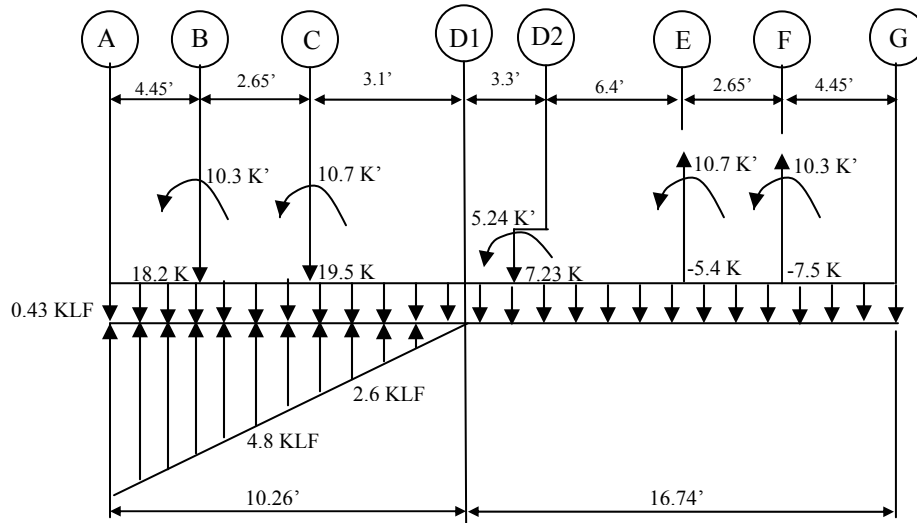
2.	$= 2.08 + \frac{(111.3)(6.4')}{245.7} = 4.98K$				
3.	$= 2.08K$				
4.	$= 1.02 - 3.03 = -2.01K$				
5.	$= 2.08 - 4.1 = -2.02K$				
Unit load =	12	14.5	5.15	-4.53	-5.51
Modification =	<u>6.18</u>	<u>4.98</u>	<u>2.08</u>	<u>-0.82</u>	<u>-2.02</u>
Final load=	18.18	19.48	7.23	-5.35	-7.53





HEATER CALCULATIONS

Sample Design 1: Cylindrical Heater Foundation





HEATER CALCULATIONS

Sample Design 1: Cylindrical Heater Foundation

Check Σ Vertical Forces

$$43.6K - 0.43KLF (27.0') - 18.2 - 19.5 - 7.23 + 5.4 + 7.5 = 0.00 \quad \text{O.K.}$$

Check Σ Moment @ centerline of mat (clockwise = positive)

$$43.6 \left(\frac{27.0}{2} - \frac{10.3}{3} \right) - 18.2 \times 9.05 - 19.5 \times 6.4 - 10.7 - 10.3 = 5.24 + 5.4 \times 6.4 + 7.5 \times 9.05 + 10.7 + 10.3$$

$$128.4K' \approx 128.67K' \dots\dots\dots \text{O.K}$$

Section	Shear (Kips)		Moment (Kip-FT)		Point
A - B	(4.80 - 0.43) (4.45) =	19.45	x 4.45 x 1/2 =	43.3	
	(8.55-4.80) (4.45)(1/2) =	8.34	x 4.45 x 2/3 =	24.74	
		27.8		68.01	
		-18.2		-10.3	B
		9.6		57.7	
			x 2.65=	25.4	
B-C	(2.79-0.43)(2.65)=	5.75	x 2.65x1/2=	7.62	
	(4.48-2.79)(2.65)(1/2)=	2.92	x 3.1x2/3=	6.03	
		18.3		96.8	
		-19.5		-10.7	C
		-1.23		86.1	
			x 3.1=	-3.81	
C-D ₁	(-0.43)(4.40)=	-1.33	x 3.1x1/2	-2.07	
	(2.79)(4.40)(1/2)=	4.03	x 3.1x2/3	8.33	
		1.47		88.55	
			x 3.3	4.9	
D ₁ -D	(-0.43)(2)=	-1.42	x 3.3x1/2	-2.34	
		0.05		91.11	
		-7.23		-5.24	D
		-7.18		85.9	
			x 6.4'=	-46.1	
D-E	(-0.43)(6.4)=	-2.75	x 6.4x1/2=	-8.80	
		-9.93		31	



HEATER CALCULATIONS

Sample Design 1: Cylindrical Heater Foundation

		5.4			-10.7	E
		-4.5			20.3	
				x 2.65' =	-11.93	
E-F	(-0.43)(2.65) =	-1.14		x 2.65' x 1/2 =	-1.51	
		-5.64			6.86	
		7.5			-10.3	F
		1.86			-3.44	
				x 4.45 =	8.28	
F-G	(-0.43)(4.45) =	-1.91		x 4.45 x 1/2 =	-4.25	
		-0.05	←	Closure Error →	-0.6	

Maximum moments (top & bottom)

$$M @ "R" = 96.8 \text{ K-FT} \quad (\text{Bottom tension})$$

$$M @ "T" = 91.1 \text{ K-FT} \quad (\text{Bottom tension})$$

$$M @ "V" = -3.44 \text{ K-FT} \quad (\text{Top tension})$$

Location of point "S"

$$-1.23 - (0.43)x + x [2.6 + (2.6 - 0.83x)] 1/2 = 0$$

$$-1.23 - 0.43x + 1.3x + 1.3x - 0.415x^2 = 0$$

$$-1.23 + 2.17x - 0.415x^2 = 0$$

$$x = \frac{2.17 \pm \sqrt{4.71 - 2.04}}{0.83} = \frac{2.17 \pm 1.63}{0.83} = 4.58 \text{ or } 0.65 \quad x = 0.65'$$

Find moment @ point "S"

$$M_s = (2.04 - 0.43)(4.45 + 2.65 + 0.65)^2 \left(\frac{1}{2} \right) + \left(\frac{1}{2} \right) \left(\frac{2}{3} \right) (6.51 \text{ KLF}) (7.75')^2$$

$$-10.3 \text{ K}' - 10.7 \text{ K}' - 3.3' (18.2 \text{ K}) - 0.65' (19.5 \text{ K})$$

$$M_s = 48.4 + 130.33 - 21 - 60.1 - 12.7 = 84.93 \text{ K} - \text{FT} \quad (\text{Bottom tension})$$



HEATER CALCULATIONS

Sample Design 1: Cylindrical Heater Foundation

Moments from Operating loads only

Maximum moment causing top tension in the center of footing is given by:

$$M_c = qD^2 \left(-0.075 \log \left(\frac{D}{D'} \right) + 0.025 - 0.025 \left(\frac{D}{D'} \right)^2 \right) D = 27.0' \quad D' = 18.1'$$

$D / D' = 27.0 / 18.1 = 1.50 < 1.60$ Therefore, Formula is applicable

Calculation of q = net soil bearing

Operating heater weight = $8 \times 66.3K + 9.5K = 539.9K$

Net weight of foundation columns above grade:	$8(7')(2.67')^2(0.150) =$	59.9K
	$(0.5')(1.167')^2(0.150) =$	0.10K
Net weight of foundation columns below grade:	$(2')(2.67')^2(0.05)(8) =$	5.70K
	$(2')(1.167')^2(0.05) =$	0.14K
	$(2')(603.9FT^2)(0.05) =$	<u>60.4K</u>
		126.2K

Net soil pressure

$$q_u = 1.4(539.9K + 126.2K) / 603.9FT^2 = 1.54KSF$$

$$M_{uc} = 1.54(27.0')[-0.075 \log(1.50) + 0.025 - 0.025/(1.50)^2] = 1122.7(-0.0132 + 0.025 - 0.011)$$

$$M_{uc} = 0.90K-FT \text{ (Top tension @ center)}$$

$$M_{up} = 1122.7(-0.0132 + 0.025 - 0.011)$$

$$M_{up} = 25.6K - FT \text{ (Bottom tension @ pier)}$$

Critical design moments for sizing rebar:

$$M_u = 80.60 K-FT \text{ (Bottom tension)}$$

$$M_u = 5.60 K-FT \text{ (Top tension)}$$

Top reinforcement

$$M_u = 5.60 K-FT$$

$$d \approx 24" - 2" - 1.5" = 20.5"$$

$$K_u = M_u / F = 5.60 / 0.42 = 9.76$$

$$F = d^2 / 1000 = 0.42$$



HEATER CALCULATIONS

Sample Design 1: Cylindrical Heater Foundation

$$a_u = 4.45 \rightarrow A_s = M_u / a_u d = 5.60 / (4.45)(20.5) = 0.061 \text{ in}^2/\text{FT}$$

$$\rho_{\min} = \frac{200}{f_y} = \frac{200}{60000} = 0.0033 \quad A_{s\min} = 0.068 \text{ in}^2 / \text{FT (Minimum)}$$

$$4/3 A_s = 4/3 (0.061) = \mathbf{0.082 \text{ in}^2/\text{FT}} \leftarrow \text{Governs}$$

Bottom reinforcement

$$M_u = 80.60 \text{ K-FT}$$

$$d \approx 24" - 3" - 1.5" = 19.5"$$

$$K_u = M_u / F = 80.60 / 0.38 = 212.1 \quad F = d^2 / 1000 = 0.38$$

$$a_u = 4.33 \rightarrow A_s = M_u / a_u d = 80.60 / (4.33)(19.5) = \mathbf{0.95 \text{ in}^2/\text{FT}}$$

Check unbalanced moment transferred to column (Reference ACI 318-05, Section 13.5.3.2)

$$\gamma_f = \frac{1}{1 + \frac{2}{3} \sqrt{\frac{b_1}{b_2}}} \quad b_1 = b_2 = 0.60$$

60% of the column moment is considered to be transferred by flexure and is resisted by flexural reinforcement within a width of $32" + 2(1.5)(24") = 104"$

$$M_u = 261.7 \text{ K-FT (From column analysis)}$$

$$F = \frac{bd^2}{12000} = \frac{(104)(19.5)^2}{12000} = 3.3 \quad a_u = 4.45$$

$$K_u = \frac{M_u}{F} = \frac{(0.6)(261.7)}{3.30} = 47.6$$

$$A_s = \frac{M_u}{a_u d} = \frac{157.0}{(4.45)(19.5)} = 1.81 \text{ in}^2$$

$$1.81 \text{ in}^2 \left(\frac{12"}{104"} \right) = 0.21 \text{ in}^2 / \text{FT}$$

$$\frac{4}{3} A_s = 1.33(0.21 \text{ in}^2 / \text{FT}) = 0.28 \text{ in}^2 / \text{FT} \quad \text{Governs}$$

Footing steel summary

$$\text{Bottom Rebar} \quad 0.95 \text{ in}^2/\text{FT} \quad \mathbf{\text{USE: \#8 @ 10" o.c.}} \quad A_s = 0.95 \text{ in}^2/\text{FT}$$



HEATER CALCULATIONS

Sample Design 1: Cylindrical Heater Foundation

Top Rebar $0.082 \text{ in}^2/\text{FT}$ **USE: #5 @ 12" o.c.** $A_s = 0.31 \text{ in}^2/\text{FT}$

Check beam shear

Operating loads only case $V_u = 24.3 \text{ K}$

$$\text{Shear inside pier circle: } V_{u1} = -\frac{qD'}{4} = -\frac{(1.54)(18.1')}{4} = 7.0 \text{ KLF}$$

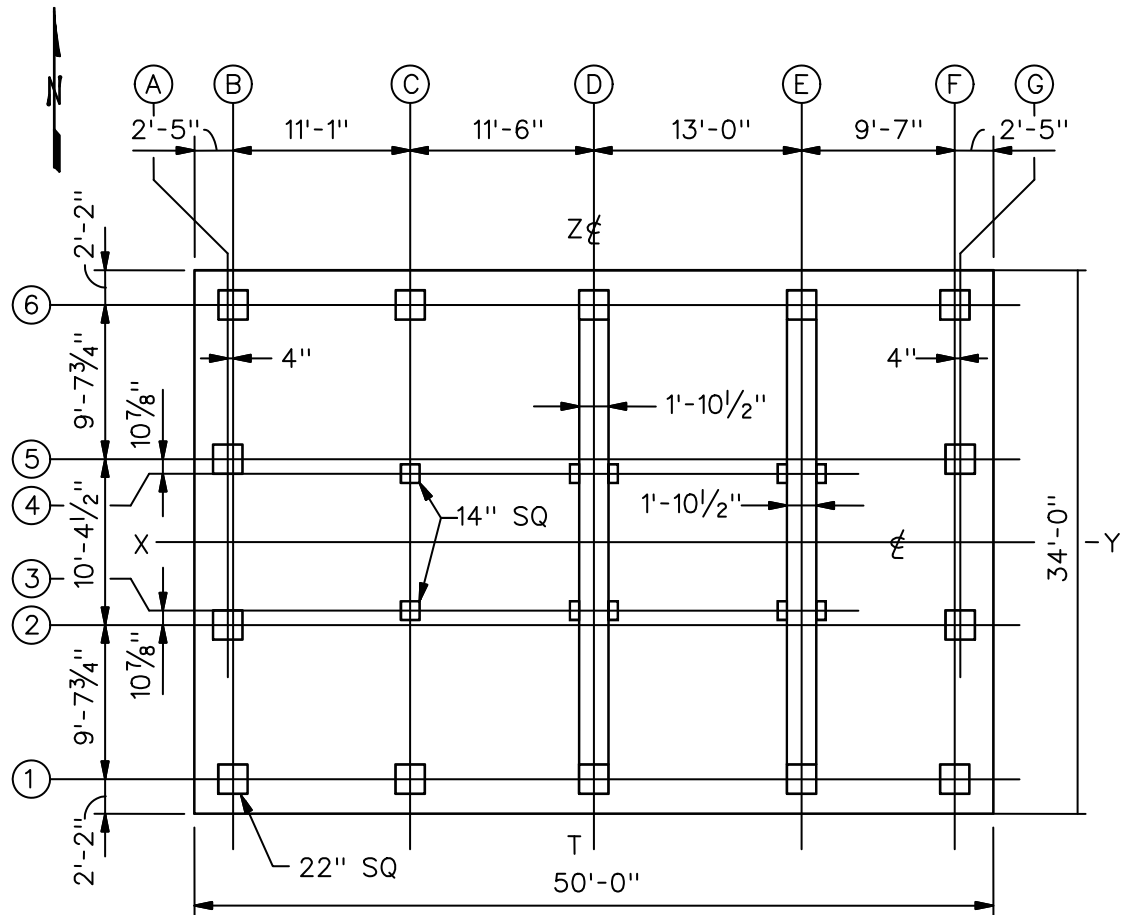
$$\begin{aligned} \text{Shear outside pier circle: } V_{u2} &= \left(-\frac{qD'}{4}\right) \left(\frac{D'}{D} - \frac{D}{D'}\right) = -\frac{(1.54)(27.0)}{4} \left(\frac{18.1}{27.0} - \frac{27.0}{18.1}\right) \\ &= -10.4 (0.67 - 1.50) = 8.60 \text{ KLF} \end{aligned}$$

$$v_u = -\frac{V_u}{0.85bd} = \frac{(24.3\text{K})}{0.85(12)(19.5)} = 0.122 \text{ KSI} \leq 2\sqrt{f'_c} = 2\sqrt{4000} = 0.126 \text{ KSI}$$

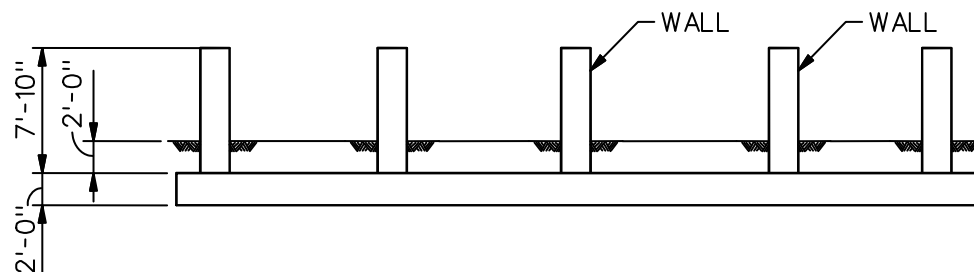
(Punching shear **O.K.** from previous analysis)

HEATER FOUNDATIONS

Sample Design 2: Rectangular Heater Foundation



PLAN



ELEVATION



HEATER FOUNDATIONS

Sample Design 2: Rectangular Heater Foundation

This is a partial sample computation to show only the mat analysis for shear and moment. Shown is the analysis in the Longitudinal (x-y) direction. In actual practice, a similar analysis is required in the transverse (z-t) direction.

Design Data:

$f'_c = 4000$ PSI $f_y = 60000$ PSI (Deformed bars)
 Net soil pressure = 4000 PSF (w/ 1/3 increase for wind or seismic)
 Unit Wt. of soil = 100 PCF
 Column loadings table (vertical & lateral) are per Vendor's "Fired Heater Loading Outline"
 Given: Seismic governs over wind Sign Convention (+↓ -↑)

Column No.	Gravity Loads		Loads @ Top of piers due to seismic @ operating condition					
	Empty Wt.	Oper. Wt.	X-Y		Z-T		@ 45°	
			V	H	V	H	V	H
B1, F1	12K	38K	+/- 23K	11K	+/- 30K	10K	+/- 32K	10K
A2, G2	18K	46K	-----	4K	+/- 11K	9K	+/- 6K	6K
A5, G5	18K	46K	-----	4K	+/- 11K	9K	+/- 6K	6K
B6, F6	12K	31K	+/- 23K	10K	+/- 30K	10K	+/- 32K	10K
C1, C6	14K	48K	+/- 13K	6K	+/- 4K	3K	+/- 7K	5K
C3, C4	3K	13K	-----	-----	-----	-----	-----	-----
D1, D6	33K	121K	-----	6K	+/- 4K	3K	+/- 7K	5K
D3, D4	4K	20K	-----	-----	-----	-----	-----	-----
E1, E6	24K	96K	+/- 13K	6K	+/- 4K	3K	+/- 7K	5K
E3, E4	3K	13K	-----	-----	-----	-----	-----	-----
Summation	141K x 2 = 282 K	472K x 2 = 944 K	47K x 2 = 94 K		47K x 2 = 94 K		47K x 2 = 94 K	

Given (cont'd): All seismic forces resisted by exterior columns only

Operating load per wall = 119 K
 Empty load per wall = 24 K
 Total operating weight = $944 + 2 \times 119 = 1182$ K
 Total empty weight = $282 + 2 \times 24 = 330$ K



HEATER FOUNDATIONS

Sample Design 2: Rectangular Heater Foundation

Solution:

Pier and foundation dead loads

Exterior Piers:	$(22 / 12)^2 \times 7.83' \times 0.150 \text{ KCF} = 3.95 \text{ K} \times 10 =$	39.5 K
Interior Piers:	$(14 / 12)^2 \times 7.83' \times 0.150 \text{ KCF} = 1.60 \text{ K} \times 2.0 =$	3.20 K
Walls:	$(22.5 / 12)^2 \times 7.83' \times 0.150 \text{ KCF} = 2.20 \text{ KLF} \times 31.5 \times 2 =$	138.6 K
Mat:	$34.0' \times 50.0' \times 2' \times 0.150 \text{ KCF} =$	510.0 K
Soil:	$2 \times (1700 - 3.36 \times 10 - 1.36 \times 2 - (1.875 \times 31.5 \times 2)) \times 0.10 =$	<u>309.1 K</u>
		1000.4 K

C.G. foundation relative to centerline (x-y direction)

$$\begin{aligned}
 69.3 \times 13.0 &= 900.9 \text{ K} \\
 2 \times 3.95 \times 11.5 &= -90.9 \text{ K} \\
 2 \times 1.60 \times 11.5 &= \underline{-36.8 \text{ K}} \\
 773.2 \text{ K} & \quad 773.2 / 1000.4 = \mathbf{0.77'}
 \end{aligned}$$

C.G. heater (operating) relative to centerline (x-y direction)

$$\begin{aligned}
 (119 + (13 + 96) \times 2) \times 13 &= 4381 \\
 (48 + 13) \times 2 \times 11.5 &= \underline{-1403} \\
 2978 \text{ K} & \quad 2978 / 1182 = \mathbf{2.52'}
 \end{aligned}$$

Combined C.G. (operating) relative to centerline (x-y direction)

$$(773.2 + 2978) / (1000.4 + 1182) = 1.70' \rightarrow (\text{of centerline})$$

Mat analysis

Analyze strip @ column line "1" in x-y direction for operating + seismic condition

$$\text{Strip width} = 2.17' + 1/2 (9.646') = \mathbf{7.00'}$$

Calculate "fictitious" soil bearing (operating + dead) w/ load factors (U = 1.2 D + 1.0 E)

$$\begin{aligned}
 \text{Operating + Dead} &= (1182 + 1000.4) \times 1.2 = \mathbf{2618.9 \text{ K} = \Sigma P} \\
 \text{Operating + Dead OTM} &= 2618.9 \times 1.7 = 4452.1 \text{ K-FT} \\
 \text{Seismic OTM} &= 23.0 \times 4 \times 22.58 = 2077.4 \\
 & \quad 13 \times 2 \times 11.5 = 299.0
 \end{aligned}$$



HEATER FOUNDATIONS

Sample Design 2: Rectangular Heater Foundation

$$\begin{aligned}
 13 \times 2 \times 13.0 &= 338.0 \\
 94 \times 9.83 &= \underline{924.0} \\
 3638.4 &= \underline{3638.4 \text{ K-FT}} \\
 &= \underline{8090.5 \text{ K-FT}} = \Sigma \text{OTM}
 \end{aligned}$$

$$SB_u = \frac{2618.9}{(34')(50')} \pm \frac{6(8090.5)}{(34')(50')^2} = 1.54 \pm 0.57$$

$$SB_u = 2.11 \times 7.0' = 14.8 \text{ K / FT}$$

$$SB_u = 0.97 \times 7.0' = 6.80 \text{ K / FT}$$

Factored loads on 7' strip

Col. Line	Dead x Load Factor	+ Seismic x Load Factor	= P _u
B	(38 + 3.95) x 1.2	+ (-23) x 1.0	= 27.3 K
C	(48 + 3.95) x 1.2	+ (-13) x 1.0	= 49.3 K
D	(121 + 119/31.5 x 5.7 + 2.2 x 5.7) x 1.2	+ (0) x 1.0	= 186.1 K
E	(96 + 119/31.5 x 5.7 + 2.2 x 5.7) x 1.2	+ (13) x 1.0	= 169.1 K
F	(38 + 3.95) x 1.2	+ (23) x 1.0	= <u>73.4 K</u>
Summation =			505.0 K

$$@ \text{ B1 \& F1 } \quad M_u = 11.0 \times 9.83 \times 1.0 = 108.1 \text{ K-FT}$$

$$@ \text{ C1, D1 \& E1 } \quad M_u = 6 \times 9.83 \times 1.0 = 59.0 \text{ K-FT}$$

$$Mat = \frac{510 + 309.1}{(34)(50)} (7.0')(1.2) = 4.05 \text{ K / FT}$$

Evaluation of unbalanced loads & moments (7' strip)

$$P = (14.8 + 6.8) \times 1/2 \times 50 - 4.05 \times 50 - 505.0 = -167.5 \text{ K} \quad \downarrow (\text{Unbalanced Load})$$

$$M = (73.4 - 27.3) \times 22.58 = 1040.9 \text{ K-FT}$$

$$169.1 \times 13 = 2198.3 \text{ K-FT}$$

$$-49.3 \times 11.5 = -566.9 \text{ K-FT}$$



HEATER FOUNDATIONS

Sample Design 2: Rectangular Heater Foundation

$$\begin{aligned} -(14.8 - 6.8) \times 50 \times 1/2 \times 50 \times 1/6 &= -1666.67 \text{ K-FT} \\ 108.1 \times 2 + 59.0 \times 3 &= \underline{393.2 \text{ K-FT}} \\ &1399 \text{ K-FT (Unbalanced Moment)} \end{aligned}$$

$$I = 2 \times 22.58^2 + 11.5^2 + 13^2 = 1321$$

$$\text{Modification } \frac{P}{N} \uparrow \pm \frac{M_c}{I} \quad \frac{M}{I} = \frac{1399}{1321} = 1.06$$

<u>Col. Line</u>	<u>P / N</u>	<u>Mc / I</u>	<u>Net</u>
B	-167.5 / 5 = -33.5	1.06 x 22.58 = 23.93	-9.57
C	-33.5	1.06 x 11.5 = 12.2	-21.3
D	-33.5		-33.5
E	-33.5	1.06 x (-13.0) = -13.8	-47.3
F	-33.5	1.06 x (-22.58) = -23.93	-57.43

Check Σ moments @ right edge of mat

		Modified P_u			
27.3 - 9.57	=	17.73	x	47.58	= 843.6
49.3 - 21.3	=	28.00	x	36.50	= 1022
186.1 - 33.5	=	152.6	x	25.00	= 3815.0
169.1 - 47.3	=	121.8	x	12.00	= 1461.6
73.4 - 57.43	=	15.97	x	2.42	= 38.64
					<u>7181.0 K-FT</u>

$$(6.8 - 4.05) \times 50 \times 50/2 = 3437.5 \text{ K-FT}$$

$$(14.8 - 6.8) \times 50/2 \times 1/3 \times 50 = 3333.3 \text{ K-FT}$$

$$\Sigma M's = \underline{393.2 \text{ K-FT}}$$

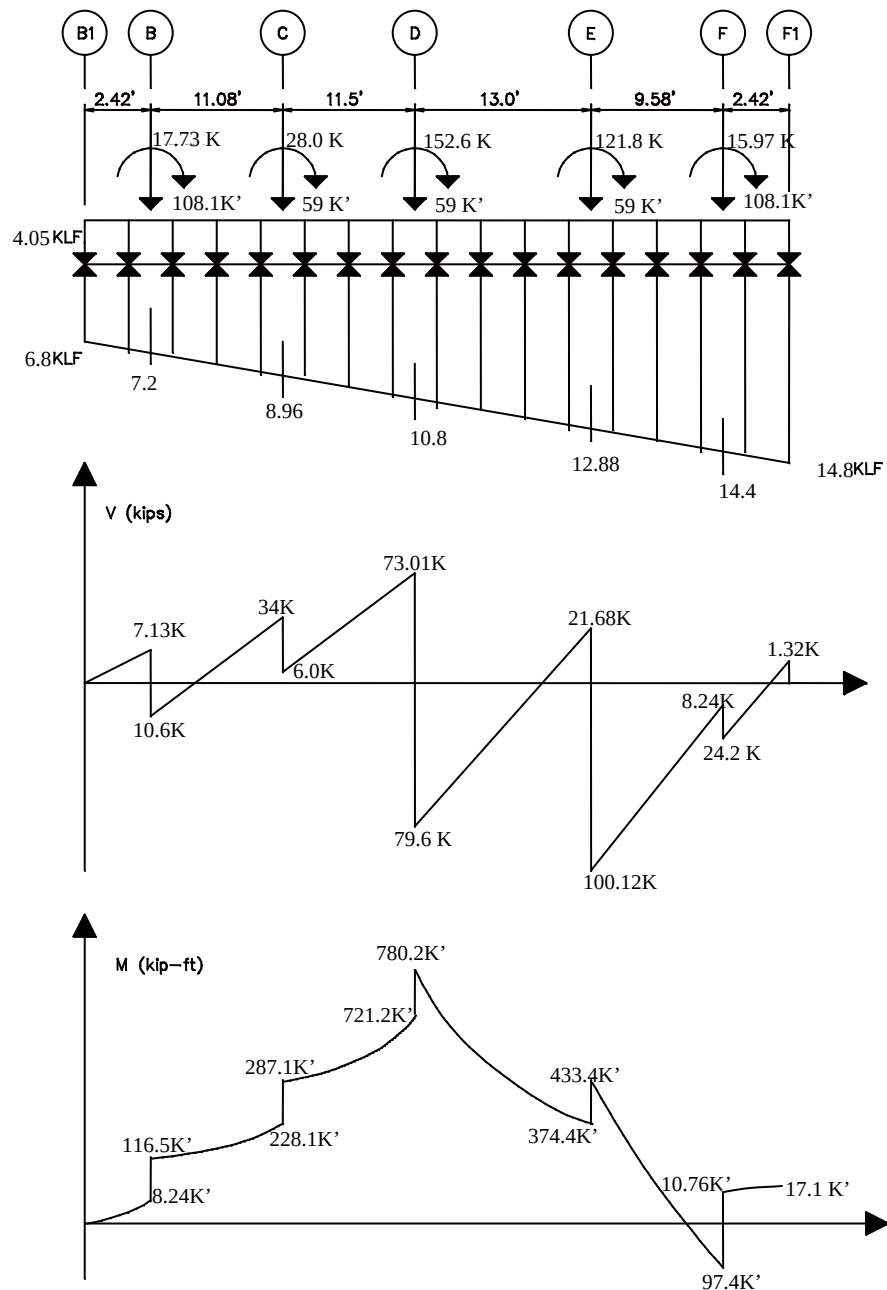
$$7164 \text{ K-FT}$$

$$\Delta = 7164 - 7181 = -17$$

$$(-17 / 7164) \times 100\% = 0.23\% < 5.0\% \quad \text{O.K.}$$

HEATER FOUNDATIONS

Sample Design 2: Rectangular Heater Foundation





HEATER FOUNDATIONS

Sample Design 2: Rectangular Heater Foundation

Section	Shear (Kips)		Moment (Kip-FT)		Point
B ₁ - B	(6.8 – 4.05) (2.42) =	6.66	x 2.42 x 1/2 =	8.06	
	(7.19 – 6.8) (2.42)(1/2) =	0.47	x 2.42 x 1/3 =	0.38	
		7.13		8.44	
		-17.73		108.1	B
		-10.6		116.54	
			x 11.08 =	-117.45	
B - C	(7.19 – 4.05)(11.08) =	34.8	x 11.08 x 1/2 =	192.8	
	(8.96 – 7.19)(11.08)(1/2) =	9.8	x 11.08 x 1/3 =	36.2	
		34		228.1	
		-28.0		59	C
		6.0		287.1	
			x 11.5 =	69	
C - D	(8.96 – 4.05) x 11.5 =	56.43	x 11.5 x 1/2 =	324.5	
	(10.8 – 8.96)(11.5)(1/2) =	10.58	x 11.5 x 1/3 =	40.6	
		73.01		721.2	
		-152.6		59	D
		-79.6		780.2	
			x 13.0 =	-1034.8	
D - E	(10.8 – 4.05) x 13.0 =	87.75	x 13.0 x 1/2 =	570.4	
	(12.88 – 10.8)(13.0)(1/2)=	13.52	x 13.0 x 1/3 =	58.6	
		21.68		374.4	
		-121.8		59	E
		-100.12		433.4	
			x 9.58 =	-959.2	
E - F	(12.88 – 4.05) x 9.58 =	84.6	x 9.58 x 1/2 =	405.2	
	(14.4 – 12.88)(9.58)(1/2)=	7.28	x 9.58 x 1/3 =	23.25	
		-8.24		-97.34	
		-15.97		108.1	F
		-24.21		10.76	
			x 2.42 =	-58.6	
F - F ₁	(14.4 – 4.05) x 2.42 =	25.05	x 2.42 x 1/2 =	30.31	
	(14.8 – 14.4)(2.42)(1/2)=	0.484	x 2.42 x 1/3 =	0.39	
		1.32	← Closure Error →	17.14	

For Design in x-y direction

+(M_U) max = 780.2 K-FT (For bottom reinforcement)

-(M_U) max = -97.4 K-FT (For top reinforcement)

(V_U) max = 100.12 K